Time Series Analysis

TSA – Cointegration and ECM model

Materials are based on dr Piotr Wójcik TSA course (QF programme)

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Outline

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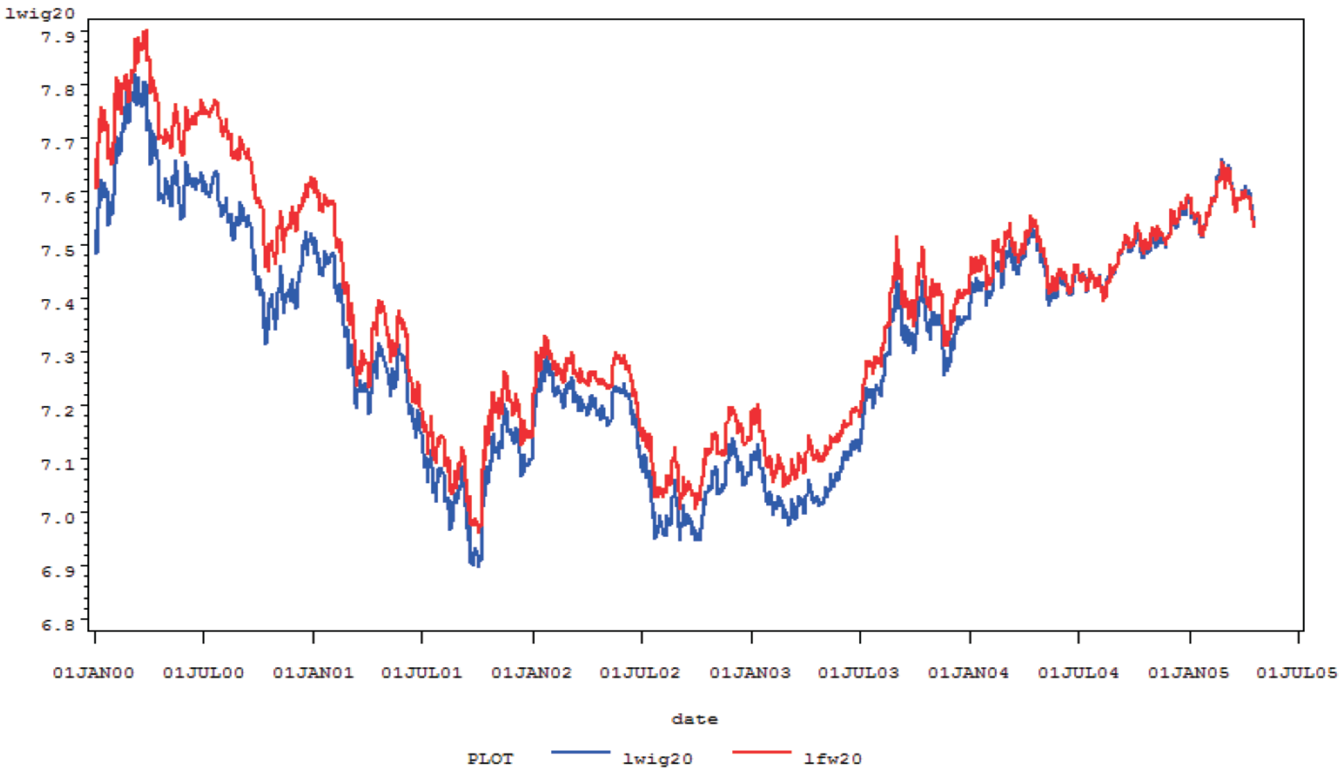
Before discussing cointegration...

- So far our conclusions refer to the fact that time series with trend are problematic:
  - spurious regressions,
  - problems with interpretation of t-statistics,
  - too high values of R^2 coefficients.
- Macroeconomic and financial time series usually contain trends.
- Possible solution could be differencing. Nevertheless, it causes loosing information about long term relationship between two or more time series.

An example of long-run relationship

- Both time series are non-stationary.
- In the same time one can observe a clear long-term relationship. Both time series seem to drift together.
- After differencing this relationship could be lost.



Cointegration of time series

Cointegration of time series

- Cointegration is a **long-run relationship** between two (or more) **non-stationary but integrated** of the same order **time series**.
  - Any **deviations** from this long-run relationship **are stationary**.
- As a result we can estimate a **model**, that simultaneously includes **short-term and long-term relationships** between two or more non stationary variables.

Definition of cointegration (Engle & Granger)

- Time series x_t and y_t are cointegrated of order d, b (where $d > b > 0$) if:
  - both time series are **integrated of order d**,
  - there exists a linear combination of these variables, $\boldsymbol{a_1 x_t + a_2 y_t}$ **(cointegration vector)**, which is integrated of **order $\boldsymbol{d - b}$**.

$$x_t, y_t \sim CI(d, b)$$

- **Cointegration** may be found **between more than two variables**.

Testing for cointegration

For two univariate time series the Engle-Granger method is usually used. It includes the following steps:

- Testing whether two time series are integrated of the same order.
- Estimation of the cointegrating vector using standard OLS regression.
- Testing whether residuals obtained in previous step are stationary.

Examples of potentially cointegrated series

- transactional prices on the spot market and forward prices for financial assetes and commodities,
- relationship between relative prices and nominal exchange rate - Purchasing Power Parity (PPP),
- relationship between price indices (PPI vs. CPI).

Cointegration of three or more variables

- If **two (or more)** variables are **integrated of different orders**, they **cannot be cointegrated**.
- Nevertheless, it is possible to **find equilibrium relationships among groups of variables that are integrated of different orders**.
- Suppose that $\boldsymbol{x_{1t}}$ **and** $\boldsymbol{x_{2t}}$ are $\boldsymbol{I(2)}$ and that other variables under consideration (i.e. $\boldsymbol{y_t}$) are $\boldsymbol{I(1)}$:

$$y_t = \beta_1 x_{1,t} + \beta_2 x_{2,t} + u_t$$

  - As such, there **cannot be a cointegrating** relationship **between** $\boldsymbol{x_{1t}}$ **(or** $\boldsymbol{x_{2t}}$) and $\boldsymbol{y_t}$.
  - However, **if** $\boldsymbol{x_{1t}}$ **and** $\boldsymbol{x_{2t}}$ **are CI(2,1)**, there exists a linear combination of the form $\boldsymbol{\beta_1 x_{1t} + \beta_2 x_{2t}}$ **which is I(1).**
  - It is possible that *this* **combination of** $\boldsymbol{x_{1t}}$ **and** $\boldsymbol{x_{2t}}$ is **cointegrated with the** $\boldsymbol{I(1)}$ **variables** and hence the error term u_t is stationary.
  - Such circumstance is referred to as **multicointegration**.
- If **three or more variables** are indeed **cointegrated**, the **Engle-Granger** method using OLS technique can find **only one cointegrating vector**, though there **can be more of them**.
- This problem can be omitted using **vector autoregression models**.

Concept of Error Correction Mechanism (ECM)

- **Cointegration of two variables** implies that there exists some **mechanism of adjustment** that **prevents the variables to deviate too far from their long-run relationship**.
  - In other words, if **two variables are cointegrated**, any **deviation from their long-run equilibrium will be stationary**.
- We can use the fact that **two non-stationary variables** are **cointegrated** to build the **model** that will **describe their long-run relationship without the risk of obtaining spurious regressions**.
- The model is expressed as:

$$\Delta y_t = \alpha_1 \Delta x_t + \alpha_2 \left(y_{t-1} - \hat{\beta}_0 - \hat{\beta}_1 x_{t-1} \right) + \varepsilon_t$$

we use the first differences of both cointegrated variables and their lagged levels as well.

- The element:

$$\alpha_2 \left(y_{t-1} - \hat{\beta}_0 - \hat{\beta}_1 x_{t-1} \right)$$

is called an **error-correction mechanism**.

- If variables $\boldsymbol{x_t}$ **&** $\boldsymbol{y_t}$ **are cointegrated** with the cointegrating vector **[1,** $\boldsymbol{-\beta_0, -\beta_1}$**]** then $\boldsymbol{\alpha_2 \left(y_{t-1} - \widehat{\beta}_0 - \right.}$

Parameter interpretation in the ECM

$$\Delta y_t = \alpha_1 \Delta x_t + \alpha_2 \left(y_{t-1} - \hat{\beta}_0 - \hat{\beta}_1 x_{t-1} \right) + \varepsilon_t$$

- $\boldsymbol{\alpha_1}$ describes the **short-term relationship** between the changes (first differences) in $\boldsymbol{x_t}$ **and** $\boldsymbol{y_t}$,
- $\boldsymbol{\alpha_2}$ describes the **speed of adjustment to the long-term relationship**.
  - it measures a **fraction of previous period's deviation from a long-run equilibrium** that will be corrected after (each) one period.
  - we expect this parameter to be **negative**, as a correction should have a reversed direction as compared with the deviation from the long-run equilibrium path.
  - $\boldsymbol{\widehat{\beta}_0}$ & $\boldsymbol{\widehat{\beta}_1}$describe the **long-run relationship between** $\boldsymbol{y_t}$ **and** $\boldsymbol{x_t}$.
- Granger's representation theorem
  - The **restrictions necessary to** ensure that the variables are **CI(1,1) guarantee that an ECM exists**.
  - This means that an **ECM for I(1)** variables **necessarily implies cointegration**.
  - So if we find that Error Correction Mechanism works for a pair of non-stationary time series, it means that they are cointegrated.
  - In other words, for any set of I(1) variables, ECM and cointegration are equivalent representations.

Estimation of ECM

Two-step **Engle-Granger's method**

- Check **integration order** for both variables of interest.
  - If **both variables are I(d) check** whether they are **cointegrated**.
  - Estimate the **cointegrating vector** and save the **residuals from OLS** regression testing for cointegration.
- Use lagged residuals from point (1):

$$\hat{u}_{t-1} = y_{t-1} - \hat{\beta}_0 - \hat{\beta}_1 x_{t-1}$$

as an independent variable in ECM equation:

$$\Delta y_t = \alpha_1 \Delta x_t + \alpha_2 \hat{u}_{t-1} + \varepsilon_t$$

Causality

- Causality is a **philosophic** rather **than an empirical issue**.
- However, in empirical economics one needs to define such a term and a method that allows to test causality empirically.
- For example, one may wish to know **whether a rise of prices causes a rise of wages or whether the process has an opposite direction**.
- Remarks about causality in economics:
  - **Future events cannot cause the past**.
  - **There is nothing like instantaneous causality**, since there is always some time between dependend actions.
  - The same applies to simultaneous bi-directional impact of two variables.
  - Nevertheless, **in practice** one needs to use the terms **„instant" and „simultaneous" causality**.
  - The reason for that is that one is **not able to measure** the levels of **variables of concern in a continuous way**. The empirical data are always collected in discrete time intervals.
- A statement: *„x influences y and simultaneously y influences x"* is some kind of simplification.
  - Since our observations are not continuous one cannot define a cause and a following effects.

Granger causality

The most popular definition of causality used in empirical economics is that proposed by Granger (1969) ``Investigating causal relations by econometric models and cross-spectral methods''. Econometrica 37 (3), 424-438.

One defines that x Granger causes y, if past values of x improve forecasts of current and future values of y (ceteris paribus).

Testing for Granger causality – algorithm:

- One selects a sensible (from the theoretical point of view) number of **lags p**
- Then one estimates a model in the form:

$$y_t = \alpha_0 + \sum_{i=1}^{p} \alpha_i y_{t-i} + \sum_{i=1}^{p} \beta_i x_{t-i} + u_t$$

- Finally, one has to test the joint hypothesis that all coefficients of $x_{t-1}, x_{t-2}, \dots, x_{t-p}$ are equal to zero:

$$H_0\colon \beta_1 = \beta_2 = \cdots = \beta_p = 0$$

- In other words, one tests the hypothesis that **x does not Granger cause y**.

Instantaneous causality

- One can easily extend the definition of Granger causality for the case of instantaneous causality.
- Instantaneous causality happens if current values of y can be better forecasted using current and past values of x rather than without them (ceteris paribus).
- The testing equation becomes:

$$y_t = \alpha_0 + \sum_{i=1}^{p} \alpha_i y_{t-i} + \sum_{i=0}^{p} \beta_i x_{t-i} + u_t$$